Candidate's Name ……

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

JOINT INSPIRING EXAMINATIONS FOR PRIVATE SECONDARY SCHOOLS

(JIEPSS)

FORM FORM ONE JOINT TERMINAL EXAMINATION

CODE 033 BIOLOGY

Time: 2:30 Hours **May/ June 2025**

Instructions

1. This paper consists of sections A, B and C with a total of **ten (10)** questions.
2. Answer **all** questions in the spaces provided.
3. Section A carries **fifteen (15)** marks, section B **seventy (70)** marks and section C carries **fifteen (15)** marks.
4. All writing should be in **blue** or **black** ink, **except** for diagrams which must be in pencil.
5. Cellular phones and any unauthorized materials are not allowed in the examination room.
6. Write your **NAME and STREAM** on every page of your answer sheet(s).

FOR EXAMINER'S USE ONLY		
QUESTION	**SCORE**	**EXAMINER'S INITIALS**
1		
2		
3		
4		
8		
9		
10		
TOTAL		

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SECTION A (15 Marks)

Answer **all** questions in this section

1. For each of the items (i) – (x), choose the correct answer from the given alternatives and write its letter in the box provided.

(i) Which statement about classification is incorrect?

A. Species is the most specific taxonomic level
B. Organisms in the same genus can interbreed freely
C. Order includes one or more families
D. Classification helps identify evolutionary links

(ii) During microscopy, damage to a high-power lens and the slide most likely results from:

A. Rotating the mirror in low light
B. Switching objectives without adjusting focus
C. Using coarse focus at high magnification
D. Cleaning lenses with cotton wool

(iii) A plant cell specialized in capturing solar energy is expected to have:

A. Thin cell wall and small vacuole
B. Irregular shape and few chloroplasts
C. Dense cytoplasm and abundant chloroplasts
D. Flat surface and few mitochondria

(iv) Which of the following would most likely challenge the universality of cell theory?

A. Discovery of a unicellular organism without genetic material
B. Observation of mitochondria reproducing independently within a cell
C. Presence of nuclei in bacterial cells
D. Existence of acellular structures showing metabolic activity

(v) Which of the following is a branch of biology?

A. Sociology
B. Agriculture
C. Anthropology
D. Botany

(vi) Which organism belong to phylum apicomplexa?

A. Paramecium
B. Plasmodium
C. Euglena
D. Amoeba

(vii) Which of the following organelles is involved in respiration?

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A. Cell membrane
B. Nucleus
C. Mitochondria
D. Lysosome

(viii) In biological investigation the aim of the experiment is to:
A. Predict the problem
B. Confirm the result
C. Identify the problem
D. Test the hypothesis

(ix) Which of the following characteristics of living things is the basic difference between plants and animals?
A. Growth
B. Movement
C. Nutrition
D. Irritability

(x) Which of the following chemical substance accelerate fire?
A. Corrosive
B. Harmful
C. Oxidant
D. Flammable

2. Match the apparatus in List A with their correct uses from List B by writing the letter of the correct response beside the item number.

LIST A	LIST B
(i) Sweep net (ii) Scoop net (iii) Pooter (iv) Quadrat (v) Mounting needle	A. Used to observe minute external features of an organism's body part. B. Used to trap ground-dwelling insects and crawling arthropods. C. Used to estimate population size and distribution of organisms in a defined area. D. Used to scoop aquatic invertebrates and organisms from water surfaces. E. Used to collect slow-moving insects from foliage using a suction technique. F. Used to catch flying insects. G. Used to lift small delicate specimens.

Answer

List A	(i)	(ii)	(iii)	(iv)	(v)
List B					

SECTION B (70 Marks)

Answer **all** questions in this section

3. (a) Draw the scientific device used for viewing very tiny specimens and label the parts using Roman numerals only for the following roles:
   i) Raises or lowers the main tube to bring image roughly into view
   ii) Brings the image into sharp detail
   iii) Bounces and directs light onto the object under view
   iv) Concentrates the light onto the slide
   v) Controls how much light passes through the slide

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(b) A school was provided with a new optical instrument for observing very small living things. After several weeks, the images became blurred and the tool stopped working properly. Suggest four proper ways of handling and maintaining such a tool to keep it functional for long.

(i) ……
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(ii) ……
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(iii) ……
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(iv) ……
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4. (a) Some living organisms are so simple that their internal contents float freely without any surrounding membranes. They reproduce rapidly and can survive in extreme environments.

(i) Identify the organisms.
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(ii) List two features that distinguish these organisms from more complex ones.

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- ……
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(b) In multicellular organisms, not all cells perform the same task. Different cells specialize to perform different tasks.

(i) What is the term for this process where cells become specialized?
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(ii) Name three different types of specialized cells and their functions.

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- ……
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- ……
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(iii) Why is this process important for survival in complex organisms?
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5. A green substance found in some plant cells allows them to trap sunlight and use it to convert carbon dioxide and water into food.

   a) Identify the biological process described.

   ……..

   b) Name the organelle where this process takes place.

   ……..

   c) Write a word equation for the process.

   d) Briefly explain how the following factors may affect the rate of the process identified in (a).

      (i) Light intensity ……...
      ……...
      ……...

      (ii) Carbon dioxide concentration
      ……...…
      ……...
      ……...

   e) "Life on earth would have been impossible without the biological process identified in (a)." Justify this statement by giving three points.

      (i) ……
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      (ii) ……
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      (iii) ……
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6. Scientists use a special process to organize living organisms based on how similar or different they are.

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a) Identify the process.

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b) Describe its two major types.

(i) ……
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(ii) ……
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c) Explain four reasons why this process is important in the study of living things.

(i) ……
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(ii) ……
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(iii) ……
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(iv) ……
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7. (a) Onion and cheek cells are different and yet similar. Justify this statement by giving three points in each case.

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(b) State any four reasons why it is more advantageous to use the name *Homo sapiens* instead of using Man when referring to Human being.

(i) ……
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(ii) ……
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(iii) ……
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(iv) ……
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8. (a) It has been established that most students don't study hard and therefore score low in biology due to lack of motivation. The students fail to see the connection between biology as an academic subject with their everyday life and their future in general. In four points, give such students an advice that could change their mindset.

(i) ……
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(ii) ……
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(iii) ……
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(iv) ……
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(b) Chinua has just graduated from UDSM where he was studying civil engineering. He has got his first job at Maji Maji secondary school where he is supposed to construct a building for carrying

out experiments. What four things he is going to consider when constructing such building? Give reason for each.

(i) ……
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(ii) ……
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(iii) ……
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(iv) ……
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9. (a) An agricultural officer advised a farmer to apply fertiliser on his farm that contains nitrogen, phosphorous and potassium (NPK). List three signs that were shown by plants that led the agricultural officer to give such advice to the farmer.

(i) ……
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(ii) ……
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(iii) ……
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(b) With two examples in each case, explain how macroelements are different from microelements.

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(c) Explain how each of the following features of the leaf make it suitable organ for photosynthesis.

(i) Petiole
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(ii) Lamina

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(iii) Midrib and veins

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SECTION C (15 Marks)

Answer question **ten (10)**

10. "Some members of the kingdom Monera are beneficial while others are harmful to humans and to the environment in general." With vivid examples where necessary, justify this statement by giving three points in each case.

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